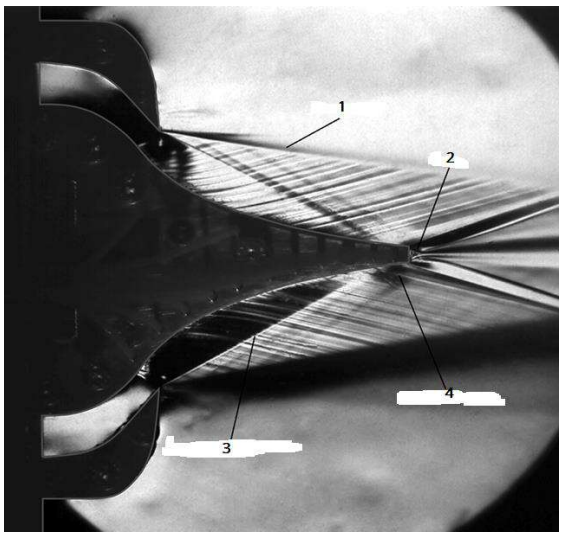
Gas Dynamics Written Test July 17th 2023 (MSAR B. Favini)

1. The schlieren photograph shows a steady two-dimensional supersonic flows for an aerospike nozzle. The expansion ratio (total pressure / external pressure) is equal to 25. Describe the flow field and identify the flow features labeled 1-2-3-4.



La fotografia Schlieren mostra un flusso supersonico bidimensionali stazionario per un ugello aerospike. Il rapporto di espansione (pressione totale / pressione esterna) è pari a 25. Descrivere il campo di flusso e identificare le caratteristiche del flusso etichettate 1-2-3-4.

(10/30)

2. A plane shock propagates inside a constant section duct. The air upstream of the shock is at rest with the pressure p=1 and the speed of sound a=1. The absolute Mach number downstream of the shock is M=1.87. The shock interacts with an abrupt reduction in area equal to 1.5. Determine the change in the shock intensity due to its interaction with the area change. Sketch the flowfield on the x-t and p-u planes.



Un urto piano propaga all'interno di un condotto a sezione costante. L'aria a monte dell'urto è in condizione di quiete con la pressione p=1 e la velocità del suono a=1. Il numero di Mach assoluto a valle dell'urto M=1.87. L'urto incontra una brusca riduzione d'area pari a 1.5. Determinare la variazione dell'intensità dell'urto a causa della sua interazione con la variazione d'area. Disegna il flusso sui piani x-t e p-u.

(20/30)

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The unsteady shock waves propagates along the tube full of quiet air (u=0). The air state is given in non dimensionalized variables. The Mach number of the state behind the shock, the grey region located on the left, is known and therefore since the state ahead, on the right of the shock, is also known by means of the jump conditions it is possible determine completely the state behind the shock.

Specifically it is possible to follow an iterative procedure.

First assume a value for the shock velocity w.

Known w compute relative Mach number of the state ahead M_{1r}=-w, since a_1=1.

Known M, absolute value since it is negative, enters the normal shock table in order to determine the state behind in terms of relative Mach number M_{2r} and sound speed a_2.

Since absolute and relative Mach numbers are related by the formula:

$M_{2r} - M_{2a} + w/a_2 = 0$

when this equation is satisfied the shock velocity w is correct otherwise it is necessary to modified its value and re-start the iterative procedure.

It is necessary to pay attention to the fact that M_{2a} is a positive quantity whereas M_{2r} on the contrary is negative.

By so doing the correct value of w is more or less 18.

The state behind the shock is more or less:

u=15 p=380 and ρ=8

Once the state behind is kown it is possible to analyzed what appens when the shock meets the abrupt area reduction.

Again by means on the tables, the isentropic one at this time, it is verified that M=1.87 has a critical section lower than the smaller tube section which means that the entire mass flow goes through the restriction.

By means of the isentropic tables it is immediate to determine the state of the flow entering the narrow duct section which is more or less:

u=12 p=1100 and ρ=18

This state, left state, together with the state ahead of the shock, right state, define a Riemann Problem whose solution is a RCS. The intermediate states are more or less:

behind the rarefaction and at the left of the contact discontinuity

u=16 p=440 and ρ=9

behind the shock and at the right of the contact discontinuity only the density is different and equal approximately 8.